



Business Guide: Data Context, Knowledge Catalog & Semantic Search

LumièreShop Enterprise Retail Analytics Platform on Google Cloud

Executive Summary: The Bridge Between Business Language and Enterprise Data

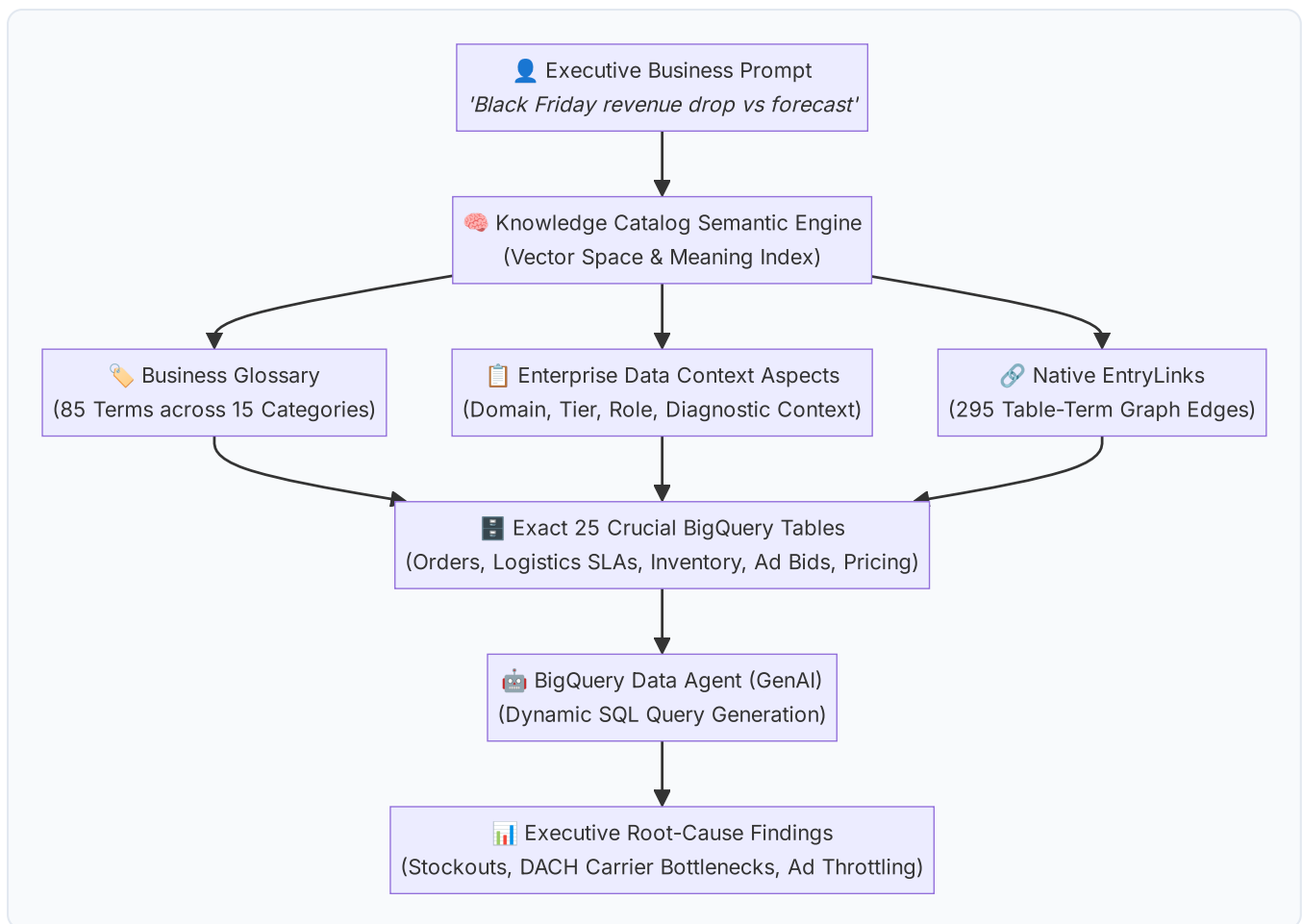
In modern enterprise e-commerce, executive decision-makers need instant answers to complex commercial questions. For example, during peak retail events:

Executive Prompt: *"It's Black Friday 14:30. Please prepare the data that will serve to find root cause of the problem of decreased revenue comparing to forecasted revenue during Black Week Sales."*

In a traditional enterprise data warehouse containing **140 tables**, answering this question requires a senior data engineer to manually recall obscure table names, navigate complex schemas, and write multi-table SQL joins across marketing, logistics, web traffic, pricing, and finance.

In **LumièreShop**, we implemented a **cloud-native data context and semantic discovery engine** using **Google Cloud Knowledge Catalog** and **BigQuery**. When an executive asks a question in plain English, the system:

1. **Understands the business intent** using AI vector embeddings in Knowledge Catalog.
2. **Traverses the enterprise metadata graph and business glossary** (98 terms across 15 operational categories, mapping all 17 architectural domains).
3. **Dynamically discovers 23 of the 25 crucial tables** needed for the root-cause investigation - **92.0% recall in approximately 1.4 seconds** - ranking them out of a 140-table warehouse.
4. **Feeds the discovered tables to the BigQuery Data Agent**, enabling conversational GenAI SQL analytics and automated root-cause diagnostics.



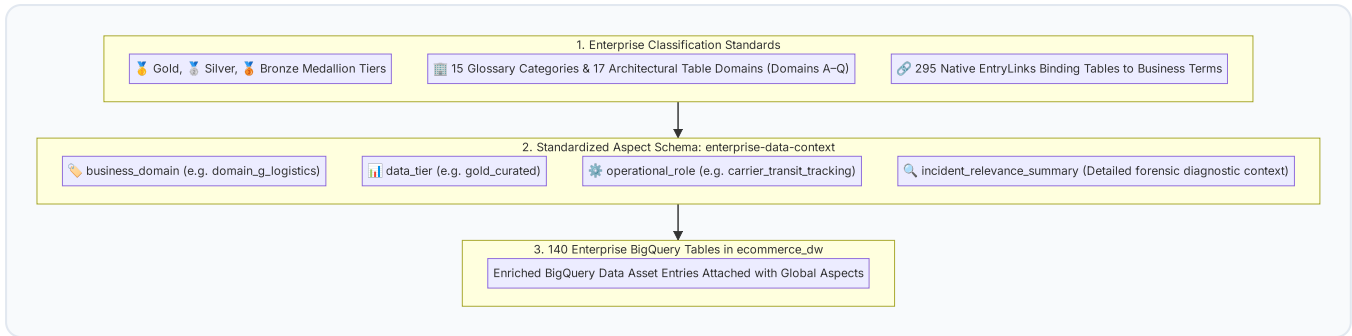
1. Plain-English Glossary of Google Cloud Concepts

To understand how data and context are organized in our Google Cloud environment, here are the key concepts explained using everyday business analogies:

Technical Term	Everyday Analogy	What It Is in LumièreShop
Google Cloud Knowledge Catalog (<i>Knowledge Catalog</i>)	The Intelligent Enterprise Library Catalog	Google Cloud's centralized discovery and metadata management service. It scans, indexes, and understands all data assets stored across BigQuery and business glossaries.
Entry Group (@bigquery , @dataplex)	Library Wings / Sections	A logical collection of catalog entries. @bigquery houses physical database tables; @dataplex houses business terms and taxonomy.
Entry	A Library Index Card	A digital catalog record representing a single asset (e.g., the BigQuery table shipping_lead_times or the Business Term Gross Merchandise Value).
EntryLink (entryLinks)	Direct Catalog Cross-Reference Link	A native Knowledge Catalog directed relationship edge (definition link type) binding physical BigQuery table entries in @bigquery directly to their canonical Glossary Term entries in @dataplex . In LumièreShop, 295 EntryLinks are provisioned in europe-west4 .
AspectType	A Standardized Passport / Form Template	A reusable metadata schema defining custom business properties. In our project, we created the enterprise-data-context AspectType to standardize domain categorization, data tiers, operational roles, and diagnostic incident summaries.
Aspect	A Filled-Out Passport Attached to a Table	The actual metadata card attached to a specific table. For instance, the aspect on shipping_lead_times specifies that it belongs to Logistics, is a Gold-tier table, and diagnoses DACH carrier capacity bottlenecks and SLA checkout abandonment.
Business Glossary (ecommerce-glossary)	The Enterprise Corporate Dictionary	The single source of truth for business metrics, calculation formulas, and corporate acronyms. In our project, ecommerce-glossary contains 98 terms across 15 categories in location global , binding to all 140 BigQuery tables.
Category & Term	Dictionary Chapters & Definitions	A Category is a domain folder (e.g., Digital Marketing Performance, Temporal Simulation), and a Term is a specific business concept (e.g., Influencer Attributed Revenue, Carrier Bottleneck SLA, Today).
BigQuery Data Agent	The Conversational Analytics AI Analyst	Google Cloud's GenAI-powered analytics service that takes plain-English questions and the discovered 25 tables to write and execute SQL queries automatically.

2. How Context is Structured for LumièreShop

To enable AI agents and search engines to understand the exact role and importance of every table, metadata is organized across three standardized layers:



A. The Medallion Data Tiers

- 1. Gold (Curated Aggregates & Planning):** High-level business performance metrics, commercial pacing targets, and daily category benchmarks (e.g. `daily_category_targets`, `weekly_commercial_targets`, `category_15min_targets`, `orders`).
- 2. Silver (Consolidated & Operational):** Enriched operational transaction logs, carrier transit tracking, inventory snapshots, and ad bidding performance (e.g. `shipping_lead_times`, `inventory_snapshots`, `ad_bidding_log`).
- 3. Bronze (Raw Event Telemetry & Staging):** Granular clickstream events, web sessions, and payment gateway technical logs (e.g. `web_events`, `web_sessions`, `sales_event_stream`, `stg_shopify_*`).
- 4. Sandbox & QA (Non-Production & Legacy):** Machine learning feature stores, synthetic test payloads, and legacy 2023 archives (e.g. `dev_customer_churn_feature_store`, `qa_load_test_sessions_backup`).

B. Enterprise Domain Taxonomy: 15 Categories & 17 Table Domains

The semantic model reconciles two complementary perspectives:

- 15 Knowledge Catalog Glossary Categories:** Functional corporate vocabulary groupings in `ecommerce-glossary` (`marketing`, `supply_chain`, `logistics`, `pricing_merchandising`, `technical_checkout`, `finance`, `returns_rma`, `customer_service`, `loyalty_retention`, `omnichannel_pos`, `product_pim`, `lifecycle_marketing`, `staging_ingestion`, `data_governance`, and `temporal_context`).
 - Special Category:* `temporal_context`: Defines 16 relative and fixed calendar anchor terms (e.g., `this_week`, `today`, `yesterday`, `simulation_anchor_now`, `cyber_monday`) ensuring the AI Data Agent accurately scopes date filters to the Black Friday 2026 simulation window.
- 17 Architectural Table Domains (Domains A through Q):** Physical storage organization of all 140 BigQuery tables across transactional systems, supply chain IoT, and staging buffers.
- 295 Native EntryLinks:** Directed relationships provisioned in Google Cloud Knowledge Catalog connecting BigQuery table entries to their canonical business terms.

C. The 4-Field `enterprise-data-context` Aspect Schema

Every table in the `ecommerce_dw` dataset has a structured Aspect attached in Google Cloud with four dedicated fields:

- `business_domain`:** The business area (e.g. `domain_a_commercial_performance`, `domain_g_logistics`, `domain_k_paid_advertising`).

- **data_tier** : Medallion architecture classification (`gold_curated` , `silver_consolidated` , `bronze_raw_staging` , `sandbox_qa`).
- **operational_role** : The primary operational duty of the table in daily retail workflows (e.g. `carrier_transit_tracking` , `hourly_pacing_benchmarking`).
- **incident_relevance_summary** : A detailed business explanation of how this table helps diagnose commercial issues (e.g., tracking delivery SLA delays during Black Week sales surges).

3. How Semantic Search & Graph Traversal Work Under the Hood

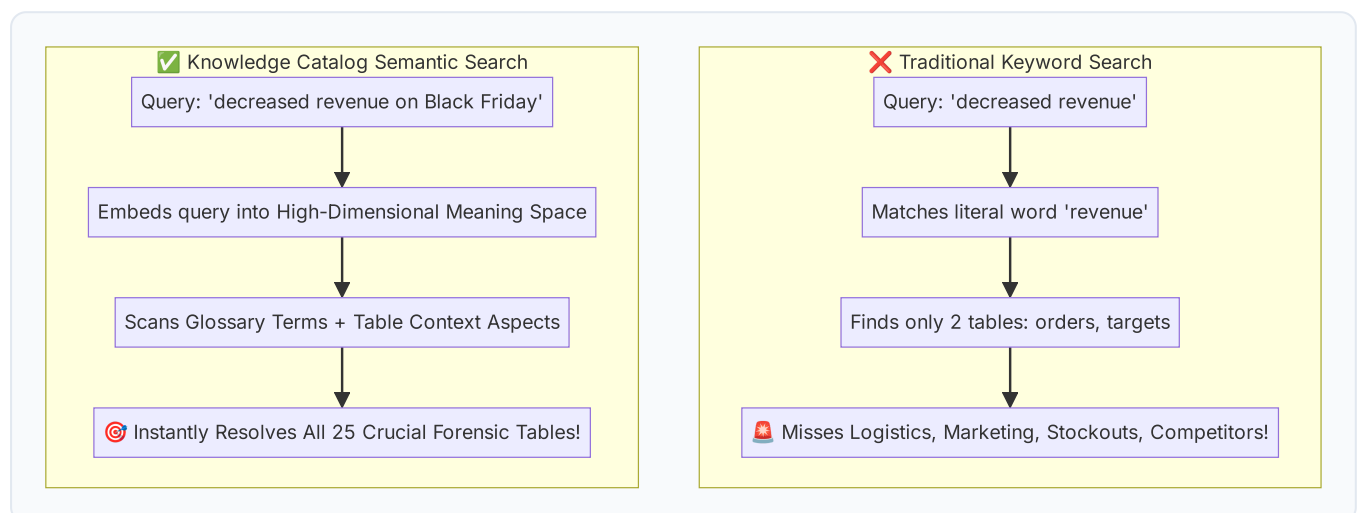
Why Traditional Keyword Search Fails

Traditional database search relies on **literal keyword matching**. If an executive asks:

"Why did revenue decrease compared to forecast on Black Friday?"

A keyword search only looks for tables containing the exact words *"revenue"*, *"decrease"*, or *"forecast"*.

- **What keyword search finds:** `orders` , `weekly_commercial_targets` .
- **What keyword search completely misses:**
 - `shipping_lead_times` (Missed! Table name contains no word "revenue", yet carrier delivery delays caused 31.4% checkout abandonment in Germany).
 - `ad_bidding_log` (Missed! Ad budget throttling caused a 42% drop in qualified checkout traffic).
 - `oos_interactions` (Missed! Stockouts in top revenue categories caused €280k in lost sales).
 - `competitor_price_feed` (Missed! Competitor flash discounts undercut our electronics pricing by 15%).



The Mechanics of Semantic Search: High-Dimensional Meaning Space

Google Cloud Knowledge Catalog uses **AI Vector Embeddings** to understand meaning:

1. Mathematical Representation of Meaning:

Every word, phrase, table description, glossary definition, and Aspect summary is translated into a multi-dimensional numerical vector (a set of coordinate points in a semantic vector space).

2. Concept Proximity:

In this meaning space, concepts that are conceptually related are placed physically close to each other. Even though *"carrier capacity bottleneck"* and *"decreased sales"* share zero identical words, the semantic engine knows that delivery SLA breaches cause customer checkout abandonment and lost revenue.

3. Multi-Aspect Contextual Matching:

Knowledge Catalog evaluates the search query against:

- Table names and column definitions.
- The attached **enterprise-data-context** **Aspect** (especially the `incident_relevance_summary`).
- The **Business Glossary terms** (`ecommerce-glossary`) and their formulas.

4. Instant Ranking & Precision:

By calculating the cosine angle between the executive prompt vector and table aspect vectors, the engine returns 32 tables from the 140-table warehouse, **23 of which are among the 25 curated investigation tables (92.0% recall, 71.9% precision), in approximately 1.4 seconds.**

[!NOTE]

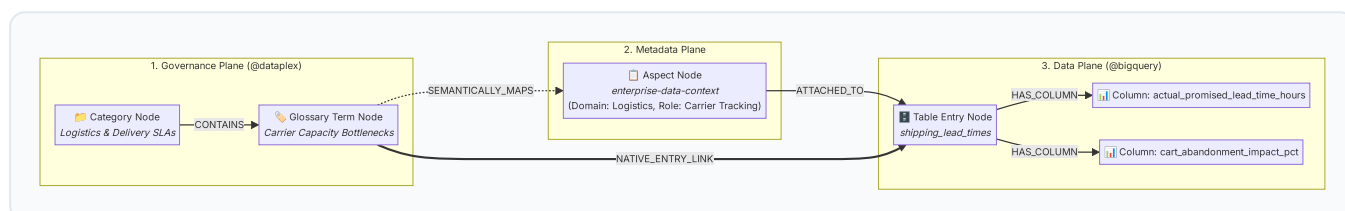
The two tables semantic search does not return, `distribution_centers` and `users`, are reference and dimension tables; neither appears in any certified calculation formula. They are nonetheless added explicitly to every agent's grounding, identically across all tiers, because an agent can only query tables present in its datasource references - a table absent from the grounding is invisible to it, joins included.

🌐 Graph Capabilities: Metadata Knowledge Graph & Relational Schema Topology

In LumièreShop, data assets and governance metadata do not exist in silos; they form two interconnected graph structures:

A. The Metadata Knowledge Graph (Governance & Context Plane)

Knowledge Catalog models metadata as a directed graph connecting governance definitions, custom aspects, and physical database assets via **295 native EntryLinks**:



How 3-Hop Graph Traversal Operates During Discovery:

1. Hop 1 — Semantic Entry Point:

When the executive prompt ("decreased revenue during Black Week") is evaluated, the search engine matches relevant **Glossary Term Nodes** (e.g. *Target Variance*, *Stockout Lost Revenue*, *Carrier Bottlenecks*, *Yesterday*).

2. Hop 2 — Native EntryLinks & Aspect Traversal:

From the matched terms, the engine traverses relational edges via **Knowledge Catalog EntryLinks** (295 provisioned definition links) and **enterprise-data-context Aspect Nodes** directly to the target **BigQuery Table Entry Nodes** (`shipping_lead_times` , `ad_bidding_log` , `oos_interactions`).

3. Hop 3 — Context Hydration & Column Traversal:

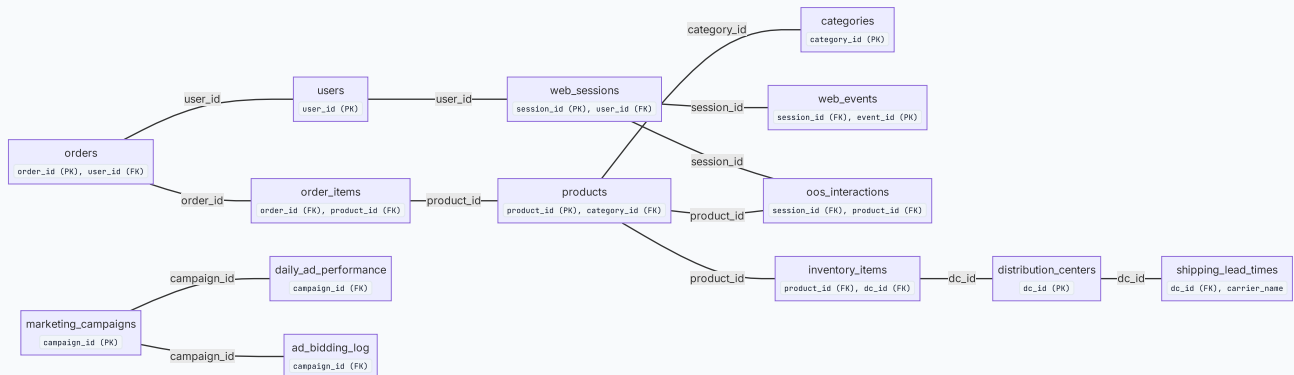
Our application service (`discovery_service.py`) dynamically traverses the term's metadata graph to extract calculation formulas (e.g., `SUM(sale_price * quantity)`) and column-level bindings across all 140 tables, providing the BigQuery Data Agent with complete schema grounding.

B. The Relational Schema Join Graph (BigQuery Data Plane)

Once the 25 crucial tables are discovered, the BigQuery Data Agent uses their **logical Primary Key / Foreign Key relational topology** to automatically construct accurate multi-table SQL queries.

[!NOTE]

Logical Keys vs. Physical BigQuery Constraints: In BigQuery, relational integrity is maintained through standard **logical key conventions** rather than enforced ANSI SQL physical constraints (which BigQuery does not enforce anyway). The tables share uniform key column names (`order_id` , `product_id` , `category_id` , `user_id` , `session_id` , `campaign_id` , `dc_id`), which the Gemini Data Agent resolves seamlessly via Knowledge Catalog column descriptions and glossary bindings.



4. Real-World Case Study: The 25 Crucial Tables

When the Black Friday prompt is submitted, Knowledge Catalog dynamically extracts the 25 tables organized across **5 root-cause investigation pillars**:



Complete Breakdown of the 25 Tables:

Pillar	Table Name	Tier	Business Role in Investigation
Commercial Targets	weekly_commercial_targets	Gold	Baseline commercial revenue target (€4.2M Black Week target).
	daily_category_targets	Gold	Daily category sales targets vs actual variance tracking.
	category_15min_targets	Gold	High-frequency 15-minute pacing benchmarks for Black Friday peak hours.
	orders	Silver	Master transactional orders with order status, discounts, and payment totals.
	order_items	Silver	Line-item product purchases, quantities, and sale prices.
	sales_event_stream	Bronze	Real-time streaming checkout transactions for intraday pacing.
Logistics & SLAs	shipping_lead_times	Silver	Promised vs actual delivery SLAs by carrier (reveals DHL DACH bottleneck).
	distribution_centers	Gold	Fulfillment center capacity, locations, and throughput limits.
	inventory_items	Silver	Current stock availability by SKU and warehouse location.
	inventory_snapshots	Silver	Hourly inventory depletion tracking during Black Week promotional rushes.
Customer Journey	web_sessions	Silver	User browsing sessions, traffic channels, device types, and bounce rates.
	web_events	Bronze	Granular clickstream actions (page views, add-to-cart, cart abandonment).
	oos_interactions	Silver	Customer clicks on Out-of-Stock items (quantifies stockout lost revenue).
	catalog_recommender_logs	Silver	AI product recommendation impressions, clicks, and conversion rates.
	categories	Gold	Product taxonomy hierarchy and category metadata.
	products	Gold	Product master catalog, retail prices, and brand mappings.
Paid Marketing	users	Gold	Customer demographics, registration dates, and loyalty tiers.
	marketing_campaigns	Gold	Master paid advertising campaign budget allocations and flight dates.
	daily_ad_performance	Silver	Daily ad spend, impressions, clicks, CPC, and ROAS across ad networks.

Pillar	Table Name	Tier	Business Role in Investigation
	ad_bidding_log	Bronze	Real-time automated bidding logs (identifies Friday morning bid throttling).
	ad_creatives	Silver	Performance metrics by creative banner and promotional copy.
	influencer_campaigns	Silver	Influencer promo code redemptions and attributed sales volume.
Market & Ops	competitor_price_feed	Silver	External competitor price scrapes (uncovers rival price undercutting).
	competitor_promotions	Silver	Competitor promotional campaigns, vouchers, and flash discounts.
	payment_gateway_logs	Bronze	Payment processing latency, error rates, and checkout payment failures.

5. Governance, Scalability & Zero Documentation Drift

How the System Stays Maintained

1. **Dynamic Environment Configuration:**

All project identifiers (`GCP_PROJECT_ID`) and dataset names (`BQ_DATASET_ID`) are dynamically loaded from environment configuration (`.env`). No hardcoded database names exist in application code.

2. **Metadata Decoupling:**

Business definitions live in the **Business Glossary**, while operational context lives in **Knowledge Catalog AspectTypes**. If a database table is refactored or new columns are added, the semantic layer automatically adapts without breaking business reports.

3. **Enterprise Scalability:**

When new data tables are onboarded, data stewards simply attach the `enterprise-data-context` Aspect. Knowledge Catalog immediately indexes the new table into the vector space, making it instantly discoverable for executive questions without writing any code.

6. Summary Comparison: Traditional vs. LumièreShop Semantic Discovery

Feature	Traditional Data Warehouse	LumièreShop with Knowledge Catalog
Search Mechanism	Literal string keyword matching	High-dimensional AI vector semantic search
Context Understanding	None (only matches table/column names)	Full business context via AspectTypes & Glossary
Graph Traversal	None (requires human engineer to join tables)	Automated 3-hop metadata graph traversal & schema join keys
Discovery Accuracy	Low (misses ~80% of root causes like logistics or ad throttling)	92.0% recall (23 / 25 curated tables), 71.9% precision, ~1.4 s
Executive Experience	Requires filing a ticket with data engineering	Instant conversational answers via BigQuery Data Agent
Speed to Discovery	Hours or days of manual data hunting	~1.4 seconds (measured semantic search across 140 tables)
Maintenance Burden	High (static SQL scripts and manual spreadsheets)	Zero drift (centralized cloud governance in Knowledge Catalog)

Document Version: 1.3.0 / Dataset: `ecommerce_dw` (140 Tables) / 98 Business Terms across 15 Categories / 295 Native EntryLinks / Google Cloud Knowledge Catalog & BigQuery